

Mathematical foundations for stochastic populations, survival and absorption

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1 Introduction

The population models developed in this thesis begin where deterministic rate equations are least reliable: with a small founding count, a compartment that may rupture, or a released cohort whose members are subsequently absorbed elsewhere. At that scale, birth and death do not merely perturb a smooth trajectory. A few early events decide whether a lineage disappears, establishes, or reaches a state from which a later catastrophe can release a substantial population. The relevant questions are consequently distributional. We need the probability of survival, the law of a population conditional on survival, and the joint distribution of counts before and after transfer between compartments.

This chapter develops the common mathematical language for those questions. It is selective rather than encyclopaedic. Conditional probability, probability generating functions (PGFs), exponential clocks and continuous-time Markov chains (CTMCs) are introduced only to the depth required by the models that follow. The same notation is then carried through discrete branching, linear birth–death dynamics, catastrophe hazards and absorption. This gives later single- and multi-type models a consistent starting point: states are integer-valued, event rates are explicit, extinction is an absorbing boundary, and model output remains separate from the assumptions used to obtain it.

Two mathematical effects recur throughout the thesis. The first is conditioning on non-extinction. A subcritical branching process dies out almost surely, and its unconditional mean decays to zero, yet the realisations that remain alive at a late time have a non-degenerate limiting law. For the binary Galton–Watson process, in which an individual divides with probability $p < \frac{1}{2}$ and otherwise dies, write

$$S_n = \mathbb{P}(Z_n > 0), \quad A(p) = \lim_{n \rightarrow \infty} \frac{S_n}{(2p)^n}.$$

The conditional mean then converges to $1/A(p)$. Thus a process may become progressively less likely to survive while the typical size of a surviving realisation approaches a positive constant. This distinction between frequency of survival and size given survival is central whenever later observations select only compartments or lineages that have persisted.

The second effect is the approach to criticality. As $p \uparrow \frac{1}{2}$, extinction remains certain on the subcritical side, but it occurs on an increasingly long time scale and the quasi-stationary mean diverges. At the critical point the survival probability has the power-law tail $S_n \sim 2/n$, so extinction is almost sure while the expected lifetime is infinite. Immediately above criticality, long-run survival has positive probability, but a small founder population can still be lost in its first few generations. The initial count therefore matters most in precisely the parameter regime where numerical convergence is slowest.

The constant $A(p)$ provides a particularly useful thread through these ideas. It has an exact infinite product, an exact series, rigorous two-sided bounds and the near-critical asymptotic

$$A(p) \sim 2(1 - 2p) \quad (p \uparrow \tfrac{1}{2}),$$

but no elementary expression is known. Its relation to the Koenigs linearisation of the logistic map explains why the iteration is structurally more complicated than its quadratic form first suggests. The result is not merely a detour into dynamical systems: $A(p)$ fixes the late conditional population size and provides a controlled way to compare discrete generations with continuous-time birth–death dynamics.

The final part of the chapter turns from survival to transfer. When a compartment ruptures, particles released into a shared medium need not enter another compartment instantaneously. An absorption period can instead be modelled as a CTMC for the counts inside and outside a receiving compartment. Bivariate PGFs compress the corresponding master equations into

first-order partial differential equations. Pure absorption and absorption with death admit direct probabilistic solutions; adding interior birth leads naturally to the method of characteristics and, in the general non-critical case, to a hypergeometric representation. This sequence supplies the analytical template used by the more structured rupture and multi-compartment models later in the thesis.

The order of development follows the argument rather than a catalogue of standard probability. [Section 2](#) establishes the PGF and CTMC machinery and solves the linear birth–death baseline. [Section 3](#) develops the binary Galton–Watson process, its extinction transition and its limiting conditional law. [Section 4](#) studies $A(p)$, including its bounds, asymptotics, dynamical interpretation and numerical evaluation. [Section 5](#) then separates the effects of founding count and catastrophe. Finally, [section 6](#) derives the absorption models and the characteristic solution required when birth, death and uptake interact. Exact identities, asymptotic statements, numerical observations and modelling approximations are labelled as such throughout; later chapters can therefore reuse the machinery without inheriting hidden assumptions.

2 Probability and Markov-process framework

The models in this thesis use two clocks. Discrete generations are appropriate when every individual is replaced at a common update, as in the Galton–Watson process. Continuous time is appropriate when births, deaths, absorptions and catastrophes occur asynchronously. Both descriptions are built from the same three operations: condition on what happens next, exploit independence where it is present, and package count distributions in generating functions.

2.1 Counts, conditioning and generating functions

All random variables are defined on a common probability space $(\Omega, \mathcal{F}, \mathbb{P})$. For events B with $\mathbb{P}(B) > 0$, conditional probability is

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Repeated use of this identity, either over a partition or at the first event of a process, gives the laws of total probability and expectation. In particular, first-step analysis is not a separate trick: it is ordinary conditioning applied at the first transition.

For a non-negative integer-valued random variable X , its PGF is

$$G_X(z) = \mathbb{E}[z^X] = \sum_{k=0}^{\infty} \mathbb{P}(X = k)z^k, \quad |z| \leq 1. \quad (1)$$

The coefficients and first two moments can be recovered by differentiation:

$$\begin{aligned} \mathbb{P}(X = k) &= \frac{G_X^{(k)}(0)}{k!}, & G_X(1) &= 1, & \mathbb{E}[X] &= G'_X(1), \\ \text{Var}(X) &= G''_X(1) + G'_X(1) - (G'_X(1))^2. \end{aligned} \quad (2)$$

The value $G_X(0) = \mathbb{P}(X = 0)$ is especially useful because state 0 represents extinction in all population models considered here. If $X = X_1 + \dots + X_N$ is a sum of independent, identically distributed counts, then

$$G_X(z) = \prod_{i=1}^N G_{X_i}(z) = G_{X_1}(z)^N. \quad (3)$$

This factorisation will account for both independent founding lineages and independent particles entering a compartment.

Joint counts require no new principle. For $\mathbf{X} = (X_1, \dots, X_d)$, define

$$G_{\mathbf{X}}(z_1, \dots, z_d) = \mathbb{E} \left[\prod_{j=1}^d z_j^{X_j} \right]. \quad (4)$$

Partial derivatives at the origin recover joint state probabilities, while derivatives at $(1, \dots, 1)$ give moments and cross-moments. The bivariate case will later track particles inside and outside a receiving compartment.

2.2 Exponential clocks and continuous-time chains

A time-homogeneous CTMC $X = (X_t)_{t \geq 0}$ on a countable state space is specified by its generator $Q = (q_{ij})$. With the row convention used here,

$$\mathbb{P}(X_{t+\Delta t} = j \mid X_t = i) = \delta_{ij} + q_{ij}\Delta t + o(\Delta t), \quad \Delta t \downarrow 0, \quad (5)$$

where $q_{ij} \geq 0$ for $j \neq i$ and $q_{ii} = -\sum_{j \neq i} q_{ij}$. Thus $q_i := -q_{ii}$ is the total rate of leaving state i . Conditional on the chain being in i , the holding time is $\text{Exp}(q_i)$, after which the next state is $j \neq i$ with probability q_{ij}/q_i . An absorbing state has $q_i = 0$; population zero is the main example.

This construction rests on the memoryless property of the exponential law. If $T \sim \text{Exp}(\lambda)$, then

$$\mathbb{P}(T > s + t \mid T > s) = \mathbb{P}(T > t) = e^{-\lambda t}.$$

For independent clocks $T_r \sim \text{Exp}(a_r)$, their minimum and its identity obey

$$\min_r T_r \sim \text{Exp}\left(\sum_r a_r\right), \quad \mathbb{P}\left(T_j = \min_r T_r\right) = \frac{a_j}{\sum_r a_r}. \quad (6)$$

The winning clock is independent of the minimum waiting time. Equation (6) is the pathwise meaning of a rate model: one may draw one holding time from the total rate and then select the event in proportion to its rate.

Let $P_{ij}(t) = \mathbb{P}(X_t = j \mid X_0 = i)$ and $P(t) = (P_{ij}(t))$. Under the usual regularity and non-explosion conditions, the transition semigroup satisfies

$$P'(t) = P(t)Q \quad (\text{forward}), \quad P'(t) = QP(t) \quad (\text{backward}), \quad P(0) = I. \quad (7)$$

Both equalities hold because $P(t) = \exp(tQ)$, but their componentwise meanings are different. If $\mathbf{p}(t)$ is a row vector of state probabilities, the forward or master equation is

$$\frac{dp_j}{dt} = \sum_i p_i(t) q_{ij}. \quad (8)$$

It records probability entering state j through all incoming transitions and leaving it through the diagonal rate. This row convention is retained in every continuous-time calculation below.

When an analytic solution is unavailable, (6) also gives an exact simulation scheme. In state x , with event propensities $a_r(x)$, draw a holding time from $\text{Exp}(\sum_r a_r(x))$, choose event r with probability $a_r(x)/\sum_k a_k(x)$, update the state, and repeat. This is the direct Gillespie algorithm. In this thesis simulation is used to check or evaluate an analytical construction, not to conceal an undefined transition mechanism.

2.3 The linear birth–death baseline

The linear birth–death process is the continuous-time reference model for the chapter. From a population of size n , births occur at total rate $n\lambda$ and deaths at total rate $n\mu$:

$$q_{n,n+1} = n\lambda, \quad q_{n,n-1} = n\mu, \quad q_{nn} = -n(\lambda + \mu), \quad (9)$$

with 0 absorbing. The per-capita rates are constant, so the mean follows the deterministic equation

$$\frac{d}{dt}\mathbb{E}[X_t] = (\lambda - \mu)\mathbb{E}[X_t], \quad \mathbb{E}[X_t | X_0 = N] = Ne^{(\lambda - \mu)t}. \quad (10)$$

The mean identifies the growth regime but not the probability of establishment; that distinction is the reason for retaining the stochastic model [1].

Write $p_n(t) = \mathbb{P}(X_t = n | X_0 = N)$. From (8),

$$p'_0(t) = \mu p_1(t), \quad (11)$$

$$p'_n(t) = \lambda(n-1)p_{n-1}(t) + \mu(n+1)p_{n+1}(t) - n(\lambda + \mu)p_n(t), \quad n \geq 1. \quad (12)$$

Solving this infinite system term by term is unattractive. Its time-dependent PGF

$$G_N(z, t) = \sum_{n=0}^{\infty} p_n(t) z^n$$

compresses the system into a single first-order PDE. Multiplying (11)–(12) by z^n , summing and shifting indices gives

$$\frac{\partial G_N}{\partial t} = (\lambda z - \mu)(z - 1) \frac{\partial G_N}{\partial z}, \quad G_N(z, 0) = z^N. \quad (13)$$

The branching property implies $G_N = g^N$, where $g = G_1$. For $\lambda \neq \mu$, direct solution of (13) gives

$$g(z, t) = \frac{\mu(1 - z) - (\mu - \lambda z)e^{-(\lambda - \mu)t}}{\lambda(1 - z) - (\mu - \lambda z)e^{-(\lambda - \mu)t}}. \quad (14)$$

At the critical value $\lambda = \mu$, the continuous limit of this expression is

$$g(z, t) = 1 - \frac{1 - z}{1 + \lambda t(1 - z)}. \quad (15)$$

Both formulae satisfy $g(z, 0) = z$, $g(1, t) = 1$, and $\partial_z g(1, t) = e^{(\lambda - \mu)t}$, which recover the initial condition, normalisation and mean.

Setting $z = 0$ yields the finite-time extinction probability quoted later:

$$\mathbb{P}(X_t = 0 | X_0 = N) = \begin{cases} \left[\frac{\mu(1 - e^{-(\lambda - \mu)t})}{\lambda - \mu e^{-(\lambda - \mu)t}} \right]^N, & \lambda \neq \mu, \\ \left(\frac{\lambda t}{1 + \lambda t} \right)^N, & \lambda = \mu. \end{cases} \quad (16)$$

Consequently the ultimate extinction probability from one founder is 1 when $\lambda \leq \mu$ and μ/λ when $\lambda > \mu$; from N independent founders it is the N th power of this value.

2.4 Continuous-time quasi-stationarity

The subcritical case $\lambda < \mu$ already exhibits the distinction between extinction and the state observed before extinction. Put $\rho = \lambda/\mu < 1$. From (14), the conditional PGF converges to

$$\frac{g(z, t) - g(0, t)}{1 - g(0, t)} \longrightarrow \frac{(1 - \rho)z}{1 - \rho z} \quad (t \rightarrow \infty). \quad (17)$$

The limiting conditional distribution is therefore geometric on $\{1, 2, \dots\}$:

$$\mathbb{P}(X_\infty^* = k) = (1 - \rho)\rho^{k-1}, \quad \mathbb{E}[X_\infty^*] = \frac{1}{1 - \rho} = \frac{\mu}{\mu - \lambda}. \quad (18)$$

The same limit is obtained from any fixed $N \geq 1$, because at a sufficiently late time the event of survival is dominated by one surviving founding lineage.

For comparison with the discrete binary model, parameterise an individual event by its probability of being a birth,

$$p = \frac{\lambda}{\lambda + \mu}. \quad (19)$$

Then $\rho = p/(1 - p)$, and the reciprocal of the limiting conditional mean is

$$A_c(p) = 1 - \rho = \frac{1 - 2p}{1 - p}. \quad (20)$$

This continuous-time constant is explicit. The corresponding discrete constant is not, and the difference between them will isolate the effect of replacing asynchronous ± 1 events by synchronous binary generations.

3 Branching, extinction and quasi-stationarity

The linear birth–death process resolves events one at a time. A Galton–Watson process instead advances by generations, replacing every member of the current population simultaneously by a random number of descendants. This change of clock preserves the branching property but alters the states available to a surviving population. That difference will eventually account for the gap between $A_c(p)$ and its discrete counterpart.

3.1 The Galton–Watson construction

Let $\{\xi_{n,i} : n \geq 0, i \geq 1\}$ be independent copies of a non-negative integer-valued random variable ξ , with

$$\mathbb{P}(\xi = k) = p_k, \quad \phi(s) = \mathbb{E}[s^\xi] = \sum_{k=0}^{\infty} p_k s^k, \quad m = \mathbb{E}[\xi] = \phi'(1).$$

Starting from one ancestor, $Z_0 = 1$, define

$$Z_{n+1} = \sum_{i=1}^{Z_n} \xi_{n,i}, \quad (21)$$

with an empty sum equal to zero. Conditional independence gives

$$\mathbb{E}[s^{Z_{n+1}} \mid Z_n] = \phi(s)^{Z_n}. \quad (22)$$

Consequently the PGF $F_n(s) = \mathbb{E}[s^{Z_n}]$ satisfies $F_{n+1} = F_n \circ \phi = \phi^{\circ(n+1)}$, while

$$\mathbb{E}[Z_{n+1} \mid Z_n] = mZ_n, \quad \mathbb{E}[Z_n] = m^n. \quad (23)$$

The extinction probability by generation n is $u_n = F_n(0) = \mathbb{P}(Z_n = 0)$. Since $F_{n+1}(0) = \phi(F_n(0))$,

$$u_{n+1} = \phi(u_n), \quad u_0 = 0. \quad (24)$$

The ultimate extinction probability is the smallest fixed point of ϕ in $[0, 1]$ under the usual non-degeneracy assumptions [2, 3]. In particular, extinction is certain when $m \leq 1$, whereas $m > 1$ permits survival with positive probability.

3.2 The multi-type extension

Later compartment models distinguish particles by type, location or phenotype. The Galton–Watson construction extends without changing its logic. Suppose there are d types and a type- i individual produces the offspring vector $\boldsymbol{\xi}^{(i)} = (\xi_1^{(i)}, \dots, \xi_d^{(i)})$, with PGF

$$\Phi_i(\mathbf{s}) = \mathbb{E} \left[\prod_{j=1}^d s_j^{\xi_j^{(i)}} \right], \quad i = 1, \dots, d. \quad (25)$$

The mean progeny matrix

$$M_{ij} = \left. \frac{\partial \Phi_i}{\partial s_j} \right|_{\mathbf{s}=\mathbf{1}} = \mathbb{E}[\xi_j^{(i)}] \quad (26)$$

records the expected number of type- j children produced by one type- i parent. If population composition is represented by a row vector $\mathbf{Z}_n$, then

$$\mathbb{E}[\mathbf{Z}_{n+1} \mid \mathbf{Z}_n] = \mathbf{Z}_n M, \quad \mathbb{E}[\mathbf{Z}_n] = \mathbb{E}[\mathbf{Z}_0] M^n. \quad (27)$$

Let $u_i^{(n)}$ be the probability that all descendants are extinct by generation n when the founder has type i . Componentwise first-step conditioning gives

$$\mathbf{u}^{(n+1)} = \boldsymbol{\Phi}(\mathbf{u}^{(n)}), \quad \mathbf{u}^{(0)} = \mathbf{0}, \quad (28)$$

and the ultimate extinction vector is the least fixed point of $\boldsymbol{\Phi}$ in $[0, 1]^d$. For a finite, irreducible and non-degenerate type system, the threshold is the spectral radius $\varrho(M)$: extinction is certain when $\varrho(M) \leq 1$, whereas $\varrho(M) > 1$ gives a positive survival probability [3]. This criterion describes reproductive amplification of a type mixture. It does not determine rupture or killing when the catastrophe rate depends on composition; that requires the transient kernel of the killed process as well as its mean matrix.

3.3 Binary death or division

The discrete model used throughout the chapter has the offspring law

$$\mathbb{P}(\xi = 2) = p, \quad \mathbb{P}(\xi = 0) = 1 - p, \quad \phi(s) = 1 - p + ps^2. \quad (29)$$

Its mean offspring number is $m = 2p$, so the process is subcritical, critical or supercritical according as $p < \frac{1}{2}$, $p = \frac{1}{2}$ or $p > \frac{1}{2}$. The mean $\mathbb{E}[Z_n] = (2p)^n$ identifies these three regimes, but it cannot distinguish a common early extinction from a rare surviving tree.

Let

$$S_n = \mathbb{P}(Z_n > 0) = 1 - u_n \quad (30)$$

denote survival to generation n . Substitution of $u_n = 1 - S_n$ into (24) yields the central recursion

$$S_{n+1} = 2pS_n - pS_n^2 = mS_n \left(1 - \frac{S_n}{2} \right), \quad S_0 = 1. \quad (31)$$

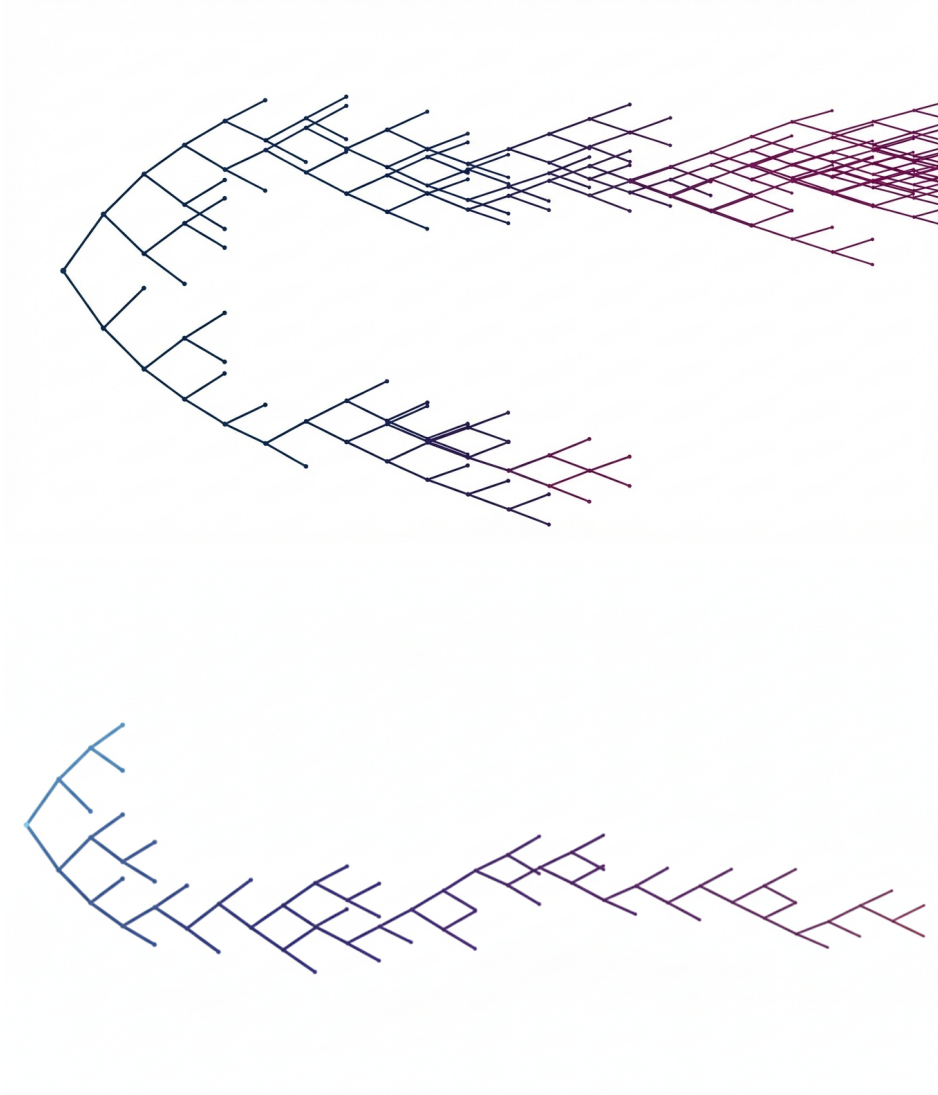


Figure 1: Genealogies of the binary Galton–Watson process, with generation increasing to the right. The upper realisation is supercritical ($p = 0.58$); the lower is subcritical ($p = 0.47$). A single trajectory does not identify the regime, but the contrast indicates why extinction and conditional size must be studied separately.

The quadratic term has a direct first-step interpretation. If the founder dies, the lineage is already extinct; if it divides, survival requires at least one of the two independent daughter lineages to remain alive.

Because extinction is absorbing, S_n is non-increasing and has a limit. The fixed points of (31) are

$$S_\infty = 0, \quad S_\infty = \frac{2p-1}{p}. \quad (32)$$

The second value is admissible only when $p > \frac{1}{2}$. Hence

$$\mathbb{P}(Z_n > 0 \text{ for all } n) = \begin{cases} 0, & 0 \leq p \leq \frac{1}{2}, \\ (2p-1)/p, & \frac{1}{2} < p \leq 1. \end{cases} \quad (33)$$

At $p = \frac{1}{2}$, the positive fixed point meets the origin and the linear contraction disappears. This is the source of the slow critical decay.

3.4 Critical survival and lifetime

At criticality, (31) becomes

$$S_{n+1} = S_n - \frac{1}{2}S_n^2. \quad (34)$$

The asymptotic form can be obtained without replacing the difference equation by an ODE. Since $S_n \downarrow 0$,

$$\frac{1}{S_{n+1}} - \frac{1}{S_n} = \frac{1}{2 - S_n} \rightarrow \frac{1}{2}. \quad (35)$$

Cesàro averaging of these increments, or the Stolz–Cesàro theorem, therefore gives

$$S_n \sim \frac{2}{n} \quad (n \rightarrow \infty). \quad (36)$$

Let $T = \inf\{n \geq 0 : Z_n = 0\}$ be the extinction generation. The tail-sum formula gives

$$\mathbb{E}[T] = \sum_{n=0}^{\infty} \mathbb{P}(T > n) = \sum_{n=0}^{\infty} S_n. \quad (37)$$

By (36), the last series has the same tail as twice the harmonic series, and hence $\mathbb{E}[T] = \infty$, although $T < \infty$ almost surely. The apparently conflicting statements are resolved by conditioning:

$$\mathbb{E}[Z_n \mid Z_n > 0] = \frac{\mathbb{E}[Z_n]}{S_n} = \frac{1}{S_n} \sim \frac{n}{2}. \quad (38)$$

Surviving critical trees become linearly larger, while their probability falls like $1/n$, leaving the unconditional mean equal to one.

Two related exponents follow from the same binary structure. First, $\mathbb{P}(T = n) = S_{n-1} - S_n = \frac{1}{2}S_{n-1}^2 \sim 2n^{-2}$. Second, if $\mathcal{T} = \sum_{j \geq 0} Z_j$ is total progeny, then a complete binary tree with k division vertices has $2k + 1$ vertices altogether. At $p = \frac{1}{2}$,

$$\mathbb{P}(\mathcal{T} = 2k + 1) = \frac{1}{2^{2k+1}} \frac{1}{k+1} \binom{2k}{k} \sim \frac{1}{2\sqrt{\pi}} k^{-3/2}. \quad (39)$$

Thus the familiar $3/2$ exponent applies along the odd support of $\mathcal{T}$, rather than to every integer indiscriminately.

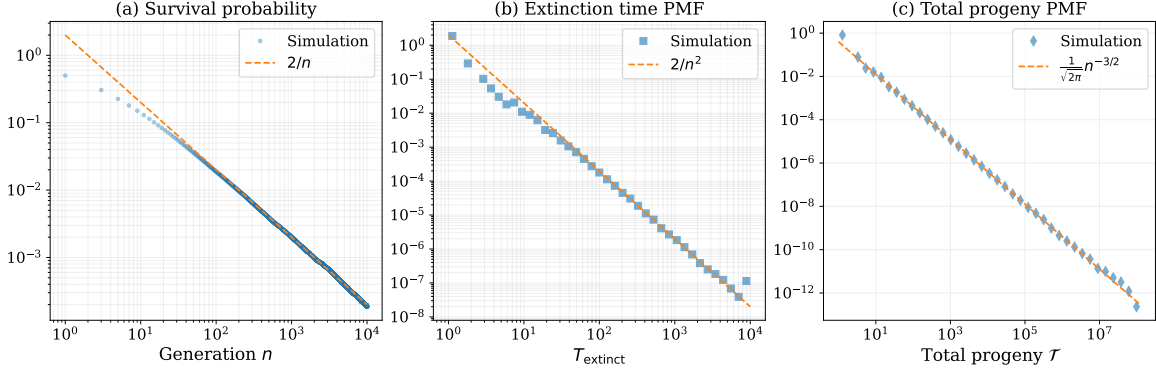


Figure 2: Critical binary branching in 10^6 simulated realisations. The panels show the n^{-1} survival tail, the n^{-2} extinction-time mass and the $n^{-3/2}$ total-progeny mass. The last scaling is understood on the odd support imposed by binary division.

3.5 The limiting conditional distribution

Return to the subcritical range $0 < p < \frac{1}{2}$, so $m = 2p < 1$. The unconditional law converges to the point mass at zero. Its informative remnant is the conditional law

$$\mathbb{P}(Z_n = k \mid Z_n > 0), \quad k \geq 1. \quad (40)$$

For a non-degenerate subcritical Galton–Watson process with the mild moment conditions satisfied here, Yaglom’s theorem states that (40) converges to a proper distribution on the positive integers [4, 3]. This limiting conditional distribution is quasi-stationary: if it is used as the initial law, one update followed by conditioning on non-extinction leaves it unchanged.

The late survival scale is obtained directly from (31). Since $S_n \rightarrow 0$, the ratio $S_{n+1}/S_n = m(1 - S_n/2)$ tends to m . Define

$$A(p) = \lim_{n \rightarrow \infty} \frac{S_n}{m^n}, \quad m = 2p. \quad (41)$$

The product argument in section 4.1 proves that this limit exists and is strictly positive for $0 < p < \frac{1}{2}$. It follows that

$$S_n \sim A(p)m^n. \quad (42)$$

The quantity on the right of (41) is a limit of ratios; it is not the limit of S_n , which remains zero.

The first conditional moment now follows without further approximation:

$$\mathbb{E}[Z_n \mid Z_n > 0] = \frac{\mathbb{E}[Z_n]}{S_n} = \frac{m^n}{S_n} \rightarrow \frac{1}{A(p)}. \quad (43)$$

The constant $1/A(p)$ will be called the quasi-stationary mean. It measures the size of a late surviving population, not the frequency with which survival occurs.

The second moment makes the same separation visible. The offspring variance is $\sigma^2 = 4p(1-p)$, and the standard branching recurrence gives, for $m \neq 1$,

$$\text{Var}(Z_n) = \sigma^2 m^{n-1} \frac{1 - m^n}{1 - m}, \quad \mathbb{E}[Z_n^2] = m^{2n} + \text{Var}(Z_n). \quad (44)$$

Dividing by $S_n \sim A(p)m^n$ and using $\sigma^2/m = 2(1-p) = 2-m$ gives

$$\mathbb{E}[Z_n^2 \mid Z_n > 0] \rightarrow \frac{1}{A(p)} \left(1 + \frac{1}{1-2p} \right), \quad (45)$$

$$\text{Var}(Z_n \mid Z_n > 0) \rightarrow \frac{1}{A(p)} \left(1 + \frac{1}{1-2p} \right) - \frac{1}{A(p)^2}. \quad (46)$$

These moment limits are consequences of the Yaglom limit, rather than a proof of distributional convergence by themselves. They show that the whole late conditional scale is controlled by p and the single constant $A(p)$.

If H denotes the PGF of the Yaglom distribution, then the branching property also yields the functional equation

$$H(\phi(s)) = mH(s) + 1 - m, \quad H(0) = 0, \quad H(1) = 1. \quad (47)$$

For the binary process its support lies in the positive even integers, because Z_n is even for every $n \geq 1$. This simple parity constraint will later provide a useful bound on $A(p)$.

3.6 A continuum approximation and its limit

Write $\varepsilon = 1 - 2p = 1 - m > 0$. The exact survival recursion can be written as

$$S_{n+1} - S_n = -\varepsilon S_n - pS_n^2. \quad (48)$$

When S_n changes little over one generation, replacing the left-hand side by a derivative gives the Riccati equation

$$\frac{dS}{dn} = -\varepsilon S - pS^2. \quad (49)$$

With $U = 1/S$, this becomes $U' = \varepsilon U + p$, and hence

$$S(n) = \frac{\varepsilon}{C\varepsilon e^{\varepsilon n} - p} \quad (50)$$

for a constant C . The approximation recovers the critical $1/n$ scale and the subcritical exponential scale, but it cannot determine their amplitude: C is fixed by the early generations, exactly where the continuum replacement is least accurate. The constant $A(p)$ therefore has to be recovered from the discrete recursion itself.

4 The discrete survival constant $A(p)$

For $0 < p < \frac{1}{2}$, every late conditional moment in [section 3.5](#) depends on

$$A(p) = \lim_{n \rightarrow \infty} \frac{S_n}{(2p)^n}.$$

The recursion for S_n makes numerical evaluation straightforward away from criticality, but it does not produce an elementary formula. This section first extracts what can be proved directly: existence, exact product and series representations, bounds and a sharp near-critical asymptotic. Only then do we turn to the question of closed form and to practical evaluation.

The endpoint $p = 0$ is excluded from the defining ratio, which would be $0/0$ after the first generation. We shall use $A(0) = \frac{1}{2}$ only for the continuous extension established below. At the other endpoint $A(\frac{1}{2}) = 0$ is likewise an extension from the subcritical interval.

4.1 Logistic iteration and an infinite product

Put

$$m = 2p, \quad w_n = \frac{S_n}{2}. \quad (51)$$

Then [\(31\)](#) becomes the logistic iteration

$$w_{n+1} = f_m(w_n), \quad f_m(w) = mw(1-w), \quad w_0 = \frac{1}{2}, \quad (52)$$

in the parameter range $0 < m < 1$. The origin is attracting throughout this range; the period-doubling behaviour of the logistic family at larger m is not involved.

Define $x_n = w_n/m^n$. From (52),

$$x_{n+1} = x_n(1 - w_n), \quad x_n = \frac{1}{2} \prod_{k=0}^{n-1} (1 - w_k). \quad (53)$$

Since $A(p) = 2 \lim_n x_n$, this gives the exact product

$$A(p) = \prod_{k=0}^{\infty} (1 - w_k) = \frac{1}{2} \prod_{k=1}^{\infty} \left(1 - \frac{S_k}{2}\right). \quad (54)$$

The convergence is not merely formal. For $0 < w_n < 1$, $w_{n+1} < mw_n$, so $w_n \leq m^n/2$ and $\sum_n w_n < \infty$. The infinite product therefore converges to a strictly positive value. This proves existence of $A(p)$ and the asymptotic $S_n \sim A(p)m^n$.

Equation (53) also corrects a possible computational misunderstanding. The raw ratios are themselves the partial products:

$$\frac{S_n}{m^n} = \prod_{k=0}^{n-1} \left(1 - \frac{S_k}{2}\right). \quad (55)$$

They decrease monotonically to $A(p)$, and hence every finite ratio is already a rigorous upper bound.

4.2 An exact series and two-sided bounds

The reciprocal constant has an additive representation that exposes the near-critical scale. Let $v_n = m^n/S_n$. Using $S_{n+1} = mS_n(1 - S_n/2)$,

$$v_{n+1} - v_n = \frac{m^n}{S_n} \left(\frac{1}{1 - S_n/2} - 1 \right) = \frac{m^n}{2 - S_n}. \quad (56)$$

Since $v_0 = 1$ and $v_n \rightarrow 1/A(p)$, summation gives the following exact identity.

Proposition 4.1 (Series representation). *For $0 < p < \frac{1}{2}$,*

$$\frac{1}{A(p)} = 1 + \sum_{n=0}^{\infty} \frac{(2p)^n}{2 - S_n}. \quad (57)$$

The survival probability lies in $(0, 1]$, so the denominator in every summand lies in $[1, 2)$. This simple observation gives bounds that remain informative at both ends of the subcritical interval.

Proposition 4.2 (Two-sided bounds). *Let $\varepsilon = 1 - 2p$. For $0 < p < \frac{1}{2}$,*

$$\frac{\varepsilon}{1 + \varepsilon} < A(p) < \frac{2\varepsilon}{1 + 2\varepsilon}, \quad (58)$$

or, in terms of p ,

$$\frac{1 - 2p}{2(1 - p)} < A(p) < \frac{2(1 - 2p)}{3 - 4p}. \quad (59)$$

In particular,

$$\frac{1}{2}A_c(p) < A(p) < A_c(p). \quad (60)$$

Proof. For every n ,

$$\frac{1}{2}m^n < \frac{m^n}{2 - S_n} \leq m^n,$$

with strictness somewhere on both sides. Substitution into (57) and summation of the geometric series yield

$$1 + \frac{1}{2\varepsilon} < \frac{1}{A(p)} < 1 + \frac{1}{\varepsilon}.$$

Inverting proves (58). The first expression in (59) is $A_c(p)/2$, while its upper bound is strictly below $A_c(p) = 2\varepsilon/(1 + \varepsilon)$. $\square$

Binary reproduction supplies a second constraint. For every $n \geq 1$, a surviving population has even size and is therefore at least two. Hence

$$\mathbb{E}[Z_n \mid Z_n > 0] \geq 2, \quad A(p) \leq \frac{1}{2}. \quad (61)$$

The inequality is strict for $p > 0$, but the lower bound in (59) tends to $1/2$ as $p \downarrow 0$. It follows by squeezing that

$$A(p) \longrightarrow \frac{1}{2} \quad (p \downarrow 0). \quad (62)$$

This limit defines the endpoint value $A(0)$.

4.3 The near-critical limit

The bounds determine the order of $A(p)$ near criticality but leave a factor of two unresolved. The exact series fixes that factor. The argument is short, but the uniform comparison it uses is important: a pointwise approximation of S_n would not by itself justify summation over n of order $1/\varepsilon$.

Theorem 4.3 (Near-critical survival amplitude). *As $p \uparrow \frac{1}{2}$,*

$$A(p) \sim 2(1 - 2p). \quad (63)$$

Equivalently, the quasi-stationary mean satisfies $1/A(p) \sim [2(1 - 2p)]^{-1}$.

Proof. Write $m = 1 - \varepsilon$, and make the dependence of the survival sequence on m explicit as $S_n(m)$. The map $mS(1 - S/2)$ is increasing in m and in $S \in [0, 1]$; induction therefore gives

$$0 \leq S_n(m) \leq S_n(1) \quad (0 < m \leq 1). \quad (64)$$

By (36), $S_n(1) \rightarrow 0$. Given $\eta > 0$, choose N such that $S_N(1) < \eta$. Monotonicity in n and (64) then imply

$$\frac{1}{2} \leq \frac{1}{2 - S_n(m)} \leq \frac{1}{2 - \eta} \quad (n \geq N).$$

Multiply (57) by ε . The initial 1 and the first N summands contribute $o(1)$, whereas $\varepsilon \sum_{n=N}^{\infty} m^n = m^N \rightarrow 1$. Thus

$$\frac{1}{2} \leq \liminf_{\varepsilon \downarrow 0} \frac{\varepsilon}{A(p)} \leq \limsup_{\varepsilon \downarrow 0} \frac{\varepsilon}{A(p)} \leq \frac{1}{2 - \eta}.$$

Letting $\eta \downarrow 0$ gives $\varepsilon/A(p) \rightarrow 1/2$, which is (63). $\square$

The continuum Riccati equation in section 3.6 reaches the same leading scale by matching its exponential tail to the critical $2/n$ decay. It is a useful interpretation, but the preceding proof shows exactly why the matching is valid. Numerically, $A(p)/[2(1 - 2p)]$ approaches one slowly. With $\varepsilon = 1 - 2p$, high-precision calculations fit the deficit as

$$1 - \frac{A(p)}{2\varepsilon} \approx \varepsilon [\log(1/\varepsilon) + 0.78]. \quad (65)$$

This is an empirical rate observation, not a proved remainder estimate.

4.4 Discrete and continuous time

The continuous-time constant from (20) is

$$A_c(p) = \frac{1 - 2p}{1 - p}.$$

At strong subcriticality, (62) gives $A(p) \rightarrow 1/2$, whereas $A_c(p) \rightarrow 1$. This factor of two is a state-space effect. A late survivor in the discrete binary process must contain at least two individuals; in the continuous-time process it may be a single individual whose death clock has not yet fired. Their limiting conditional means therefore approach two and one, respectively.

Near criticality, by contrast,

$$A_c(p) = \frac{2\varepsilon}{1 + \varepsilon} \sim 2\varepsilon \sim A(p).$$

Consequently

$$\frac{A(p)}{A_c(p)} \longrightarrow \frac{1}{2} \quad (p \downarrow 0), \quad \frac{A(p)}{A_c(p)} \longrightarrow 1 \quad (p \uparrow \tfrac{1}{2}). \quad (66)$$

Numerical values increase smoothly between these limits, but no monotonicity theorem is needed or claimed here. The rigorous comparison throughout the interval is (60).

4.5 Closed forms and the Koenigs function

Several direct searches suggest simple approximations to $A(p)$. For example, a rational symbolic-regression fit obtained from high-precision values was

$$\hat{A}_1(p) = \frac{\frac{1}{2}(1 - 2p)(1 + 2p)}{1 + \frac{1}{2}p - \frac{5}{2}p^2 - \frac{6}{5}p^3 + \frac{6}{5}p^4}. \quad (67)$$

It passes through the two endpoint values, yet

$$\frac{\hat{A}_1(p)}{2(1 - 2p)} \longrightarrow \frac{10}{11} \quad (p \uparrow \tfrac{1}{2}),$$

rather than the limit 1 required by (63). Broader tree-based searches over elementary functions produce many similarly convincing curves and the same failure of exact constraints. Near criticality, an error that is visually negligible on a plot of A can be substantial after inversion, so curve fit alone cannot settle the closed-form question.

The structural object behind $A(p)$ is the linearising coordinate of the logistic map. Let $0 < m < 1$ and $f_m(w) = mw(1 - w)$. Koenigs' theorem [5] gives a unique analytic germ ψ_m at the attracting fixed point 0 such that

$$\psi_m(f_m(w)) = m\psi_m(w), \quad \psi_m(0) = 0, \quad \psi'_m(0) = 1. \quad (68)$$

Although this definition is local, the orbit of $w_0 = 1/2$ eventually enters the local neighbourhood. Equation (68) therefore defines the required value at w_0 by continuation backwards along that orbit; no claim of global univalence is required.

Proposition 4.4 (Koenigs representation). *For $0 < p < \frac{1}{2}$, with $m = 2p$,*

$$A(p) = 2\psi_m\left(\frac{1}{2}\right). \quad (69)$$

Proof. Let $w_n = f_m^{\circ n}(1/2)$. Iterating (68) gives

$$\psi_m\left(\frac{1}{2}\right) = \frac{\psi_m(w_n)}{m^n}.$$

As $n \rightarrow \infty$, $w_n \rightarrow 0$ and the normalisation at the origin gives $\psi_m(w_n)/w_n \rightarrow 1$. Hence

$$\psi_m\left(\frac{1}{2}\right) = \lim_{n \rightarrow \infty} \frac{w_n}{m^n} = \frac{A(p)}{2}.$$

□

This representation also locates the limit of elementary methods. A theorem of Becker and Bergweiler classifies differentially algebraic solutions of Schröder, Böttcher and Abel equations for rational maps. In the Schröder case, a differentially algebraic linearising function can occur only at a repelling fixed point and then belongs to a restricted family of monomial, Chebyshev or elliptic types [6]. The fixed point here is attracting, because $0 < m < 1$. Differential algebraicity is preserved under local functional inversion, so the choice of conjugacy convention does not change the conclusion: $w \mapsto \psi_m(w)$ is hypertranscendental, and in particular non-elementary, for every fixed parameter in the branching range.

The familiar solvable logistic parameters lie outside that range. At $m = 2$, the change $u = 1 - 2w$ converts the iteration into $u \mapsto u^2$, giving

$$\psi_2(w) = -\frac{1}{2} \log(1 - 2w). \quad (70)$$

At $m = 4$, writing $w = \sin^2 \theta$ turns the update into angle doubling and gives

$$\psi_4(w) = (\arcsin \sqrt{w})^2. \quad (71)$$

Both concern a repelling origin and serve only as contrasts with $0 < m < 1$.

The affine coordinate $z = m/2 - mw$ conjugates the logistic family to

$$z \mapsto z^2 + c, \quad c = \frac{2m - m^2}{4}. \quad (72)$$

Hence the branching interval $m \in (0, 1)$ maps to $c \in (0, 1/4)$, the real centre-to-cusp radius of the main cardioid of the Mandelbrot set [7]. The parameter map gives useful dynamical context, but it does not strengthen the closed-form conclusion.

One logical boundary is essential. Hypertranscendence has been established for the function $w \mapsto \psi_m(w)$ with m fixed. It does not prove that the parameter function $p \mapsto 2\psi_{2p}(1/2) = A(p)$ has no elementary expression. The supported conclusion is narrower: no such expression is known; fixed-parameter linearisation is non-elementary; and numerical searches are consistent with, but cannot prove, the absence of a simple parameter formula.

4.6 Certified numerical evaluation

The identity (55) gives a stable calculation and a simple error certificate. Let

$$R_n = \frac{S_n}{m^n} = \prod_{k=0}^{n-1} \left(1 - \frac{S_k}{2}\right).$$

Because $S_{k+1} < mS_k$, $\sum_{k=n}^{\infty} S_k/2 \leq S_n/[2(1-m)]$. The elementary product inequality $\prod_k (1 - a_k) \geq 1 - \sum_k a_k$ then gives, whenever the lower endpoint is positive,

$$R_n \left(1 - \frac{S_n}{2(1-m)}\right) \leq A(p) \leq R_n. \quad (73)$$

Table 1: Representative values obtained from the monotone ratios and the certificate (73). The increasing cost near $p = 1/2$ reflects the critical slowdown; the displayed iteration counts use a tail target of approximately 10^{-45} , not a claim of sixty-digit accuracy.

p	$A(p)$	$1/A(p)$	iterations
0.05	0.486180336821	2.05684994695	45
0.20	0.423894537870	2.35907734274	112
0.30	0.353967177237	2.82512070132	202
0.40	0.237646658970	4.20792787214	463
0.45	0.145068746263	6.89328353462	981
0.49	0.036394897530	27.4763790497	5 125
0.499	0.003944476201	253.519085680	51 750
0.4999	0.000399257804	2504.64734704	518 021

Thus iteration may stop when the width of this interval meets the required tolerance. Accumulating logarithms of the factors avoids underflow.

For exploratory work extremely close to criticality, $2(1 - 2p)$ is an inexpensive leading approximation. It should not be joined to the product at an arbitrary fixed crossover when precision matters: the empirical logarithmic correction makes its relative error decay slowly. A reliable hybrid scheme uses (73) wherever iteration is feasible and switches to the asymptotic only after its error has been calibrated at the required parameter and tolerance.

5 Founding populations and catastrophe

Criticality controls long-time behaviour, but a small population experiences most of its immediate extinction risk before that asymptotic distinction is visible. At $p = \frac{1}{2}$, for example, the extinction probabilities in the first four generations are

$$u_1 = 0.500, \quad u_2 = 0.625, \quad u_3 = 0.695, \quad u_4 = 0.742 \quad (\text{three significant figures}). \quad (74)$$

Survival to generation two has probability $1 - u_2 = 0.375$: the founder must divide, and at least one of its two daughters must then divide. Small movements of p around one half have little time to accumulate over these first updates, so early extinction remains substantial on both sides of the critical point.

5.1 A cohort of k founders

Suppose $Z_0 = k$. The descendants of the k founders form independent lineages. If u_n is the extinction probability by generation n from one founder, all k lineages are extinct with probability u_n^k , and therefore

$$S_n^{(k)} = \mathbb{P}(Z_n > 0 \mid Z_0 = k) = 1 - u_n^k. \quad (75)$$

This formula separates the effect of founding count from the dynamics of one lineage. Taking the long-time limit gives

$$S_\infty^{(k)} = \begin{cases} 0, & 0 \leq p \leq \frac{1}{2}, \\ 1 - \left(\frac{1-p}{p}\right)^k, & \frac{1}{2} < p \leq 1. \end{cases} \quad (76)$$

A larger cohort cannot prevent eventual extinction at or below criticality. In the supercritical range it instead supplies repeated independent opportunities for one lineage to establish.

The finite-time effect is also explicit. For fixed k , expansion of $1 - (1 - S_n)^k$ at small S_n gives

$$S_n^{(k)} \sim \begin{cases} kA(p)(2p)^n, & p < \frac{1}{2}, \\ 2k/n, & p = \frac{1}{2}. \end{cases} \quad (77)$$

The cohort multiplies the late survival probability by k , to leading order, but it does not change the subcritical Yaglom scale. Indeed,

$$\mathbb{E}[Z_n \mid Z_n > 0, Z_0 = k] = \frac{k(2p)^n}{S_n^{(k)}} \longrightarrow \frac{1}{A(p)}. \quad (78)$$

At late times, conditioning selects a surviving lineage; the number of failed founders no longer affects the conditional population size.

Conditioning on survival can also bias the history that is reconstructed from successful populations. Budd and Mann’s analysis of phylogenetic clades shows that clades retained because they survive to the present are enriched for high early origination rates even under homogeneous birth–death dynamics [8]. This “push of the past” is related to (78): selection acts on realised trajectories, so an apparently exceptional early expansion need not imply a change in the underlying per-capita rates. The parallel is a modelling warning rather than an assertion that phylogenetic and within-compartment mechanisms are the same.

5.2 A discrete catastrophe mechanism

The later compartment models add a second absorbing outcome: rupture or catastrophe. A minimal discrete construction is useful because it distinguishes three quantities that are otherwise easily conflated: the hazard of catastrophe at the next update, the probability of catastrophe at an exact update, and the probability of remaining both unruptured and non-extinct.

Let each individual present at the beginning of a generation trigger catastrophe independently with probability $\kappa \in [0, 1]$, and put $c = 1 - \kappa$. Conditional on a population of size z , the probability that no individual triggers catastrophe is c^z . If catastrophe is avoided, each individual then follows the binary offspring law (29); catastrophe terminates the process.

First take pure division, $p = 1$. Conditional on reaching update j , the population has size 2^j , so the catastrophe hazard is

$$h_j = 1 - c^{2^j}. \quad (79)$$

Avoiding catastrophe for n complete updates requires every individual in generations $0, \dots, n - 1$ to avoid triggering it, and hence

$$C_n = c^{\sum_{j=0}^{n-1} 2^j} = c^{2^n - 1}. \quad (80)$$

The unconditional probability that catastrophe first occurs at update j is therefore

$$\mathbb{P}(K = j) = C_j h_j = c^{2^j - 1} (1 - c^{2^j}). \quad (81)$$

The factor C_j is indispensable: h_j alone is a conditional hazard, not an exact-time probability.

When death is restored, the census before an update is random. The process can still be described exactly by a weighted PGF. For an underlying binary branching tree starting from one individual, define

$$F_n(s) = \mathbb{E} \left[s^{Z_n} c^{\sum_{j=0}^{n-1} Z_j} \right], \quad F_0(s) = s. \quad (82)$$

Conditioning on the founder’s reproduction gives the closed recursion

$$F_{n+1}(s) = c \phi(F_n(s)) = c[1 - p + pF_n(s)^2]. \quad (83)$$

Its evaluations have distinct meanings:

$$F_n(1) = \mathbb{P}(\text{no catastrophe in the first } n \text{ updates}), \quad (84)$$

$$F_n(1) - F_n(0) = \mathbb{P}(Z_n > 0 \text{ and no catastrophe in the first } n \text{ updates}). \quad (85)$$

For $p = 1$, (83) reduces to (80); for $\kappa = 0$, it reduces to the ordinary Galton–Watson iteration.

A tempting approximation replaces the accumulated random census in (82) by its mean:

$$F_n(1) \approx c^{\sum_{j=0}^{n-1} \mathbb{E}[Z_j]} = c^{\sum_{j=0}^{n-1} (2p)^j}. \quad (86)$$

For $0 < c \leq 1$, the map $x \mapsto c^x$ is convex, so Jensen’s inequality shows that the right-hand side is a lower bound for *non-catastrophe alone*. It is not a lower bound for (85), because extinction then contributes to $F_n(1)$ but not to survival. The exact recursion (83) is cheap enough that the approximation is not needed here; its value is to show why a mean census cannot generally be inserted into a path-dependent catastrophe law without specifying the event being bounded.

Whether a process conditioned jointly on non-extinction and non-catastrophe has a quasi-stationary limit is a separate spectral question. No such limit follows from the Galton–Watson result alone, because the factor c^{Z_j} reweights the population paths. Later catastrophe models define their absorbing states and conditioning events explicitly before making that inference.

6 Absorption models and the method of characteristics

When one compartment ruptures, its contents enter a shared medium before they are taken up by the compartments that remain. Treating this transfer as instantaneous suppresses both the absorption time and any loss that occurs after uptake. The models below retain an explicit absorption period. They follow X_t , the number of particles inside one receiving compartment, and Y_t , the number still outside, from the initial state

$$X_0 = 0, \quad Y_0 = N. \quad (87)$$

The three models are deliberately ladderred. Pure absorption establishes the bivariate PGF. Interior death adds a third fate but leaves a finite state space. Interior birth then makes the state space infinite and turns the PGF equation into the non-trivial characteristic problem required by later compartment models.

For all three models write

$$p_{ij}(t) = \mathbb{P}(X_t = i, Y_t = j), \quad G(x, y, t) = \sum_{i,j \geq 0} p_{ij}(t) x^i y^j = \mathbb{E}[x^{X_t} y^{Y_t}]. \quad (88)$$

Probabilities with a negative index are understood to be zero.

6.1 Absorption only

Suppose each exterior particle is absorbed independently at rate $\alpha > 0$. From state (i, j) ,

$$\begin{aligned} \mathbb{P}(\Delta X = 1, \Delta Y = -1 \mid X_t = i, Y_t = j) &= \alpha j \Delta t + o(\Delta t), \\ \mathbb{P}(\Delta X = 0, \Delta Y = 0 \mid X_t = i, Y_t = j) &= 1 - \alpha j \Delta t + o(\Delta t). \end{aligned} \quad (89)$$

The forward equations are therefore

$$\frac{dp_{ij}}{dt} = \alpha(j+1)p_{i-1,j+1} - \alpha j p_{ij}. \quad (90)$$

Multiplying by $x^i y^j$, summing and shifting the incoming indices gives

$$\frac{\partial G}{\partial t} = \alpha(x - y) \frac{\partial G}{\partial y}, \quad G(x, y, 0) = y^N. \quad (91)$$

The probabilistic solution is immediate. One initial particle is still outside at time t with probability $e^{-\alpha t}$ and has been absorbed with the complementary probability. Its PGF is therefore

$$g_1(x, y, t) = (1 - e^{-\alpha t})x + e^{-\alpha t}y.$$

The N initial particles act independently, so

$$G(x, y, t) = \left[(1 - e^{-\alpha t})x + e^{-\alpha t}y \right]^N. \quad (92)$$

It follows that $X_t \sim \text{Binomial}(N, 1 - e^{-\alpha t})$ and $Y_t = N - X_t$. Substitution verifies that (92) also solves the PDE (91).

6.2 Absorption with interior death

Now let each absorbed particle die at rate $\mu \geq 0$. The additional transition $(i, j) \rightarrow (i - 1, j)$ has rate μi , so

$$\frac{dp_{ij}}{dt} = \alpha(j + 1)p_{i-1,j+1} + \mu(i + 1)p_{i+1,j} - (\alpha j + \mu i)p_{ij}. \quad (93)$$

The bivariate PGF satisfies

$$\frac{\partial G}{\partial t} = \mu(1 - x) \frac{\partial G}{\partial x} + \alpha(x - y) \frac{\partial G}{\partial y}, \quad G(x, y, 0) = y^N. \quad (94)$$

For a single initial particle there are three possible states at time t : still outside, alive inside, or dead. Their probabilities are

$$q_o(t) = e^{-\alpha t}, \quad (95)$$

$$q_i(t) = \begin{cases} \frac{\alpha}{\mu - \alpha} (e^{-\alpha t} - e^{-\mu t}), & \mu \neq \alpha, \\ \alpha t e^{-\alpha t}, & \mu = \alpha, \end{cases} \quad (96)$$

$$q_d(t) = 1 - q_o(t) - q_i(t). \quad (97)$$

The second line is the convolution of absorption at time s with survival inside for the remaining time $t - s$; its equal-rate expression is the regular limit omitted by a quotient-only formula. Independence now gives the trinomial PGF

$$G(x, y, t) = [q_d(t) + q_i(t)x + q_o(t)y]^N. \quad (98)$$

6.3 Characteristics on the finite-state model

The direct solution (98) provides a controlled test of the method of characteristics. Write (94) as

$$-G_t + \mu(1 - x)G_x + \alpha(x - y)G_y = 0$$

and introduce characteristic coordinates (ξ, η, τ) with initial surface

$$x(0) = \xi, \quad y(0) = \eta, \quad t(0) = 0. \quad (99)$$

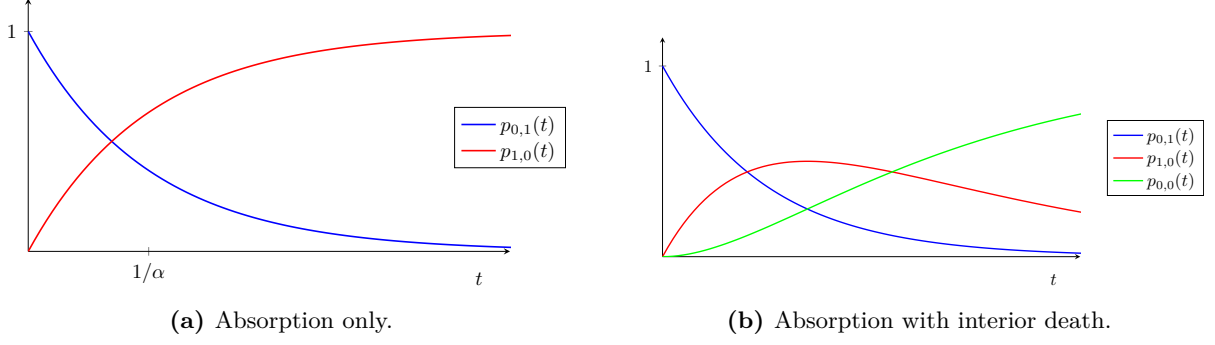


Figure 3: Single-particle state probabilities. Pure absorption transfers mass from the exterior to the interior. Adding death makes the interior probability rise and then fall as mass is transferred to the absorbing dead state.

Along a characteristic,

$$\frac{dG}{d\tau} = 0, \quad \frac{dt}{d\tau} = -1, \quad \frac{dx}{d\tau} = \mu(1-x), \quad \frac{dy}{d\tau} = \alpha(x-y). \quad (100)$$

The first two equations give $G = \eta^N$ and $t = -\tau$. The third gives

$$1-x = (1-\xi)e^{-\mu\tau}. \quad (101)$$

Substitution into the y -equation, followed by one integrating factor, yields for $\alpha \neq \mu$

$$y = 1 - \frac{\alpha}{\alpha - \mu}(1-\xi)e^{-\mu\tau} + \left[\eta + \frac{\alpha}{\alpha - \mu}(1-\xi) - 1 \right] e^{-\alpha\tau}. \quad (102)$$

Eliminating ξ and τ and solving for the initial coordinate η gives

$$\eta = q_d(t) + q_i(t)x + q_o(t)y.$$

Since $G = \eta^N$, the characteristic calculation reproduces (98). The equal-rate case follows by taking the limit, giving (96); the machinery is therefore fixed before the infinite-state model is introduced.

6.4 Absorption with interior birth and death

Finally let each interior particle give birth at rate $\beta \geq 0$ and die at rate $\mu \geq 0$. The three transitions are

$$\begin{aligned} (i, j) &\longrightarrow (i+1, j-1) && \text{at rate } \alpha j, \\ (i, j) &\longrightarrow (i+1, j) && \text{at rate } \beta i, \\ (i, j) &\longrightarrow (i-1, j) && \text{at rate } \mu i. \end{aligned}$$

Their forward equation is

$$\begin{aligned} \frac{dp_{ij}}{dt} = & \alpha(j+1)p_{i-1,j+1} + \mu(i+1)p_{i+1,j} + \beta(i-1)p_{i-1,j} \\ & - (\alpha j + \mu i + \beta i)p_{ij}, \end{aligned} \quad (103)$$

and the PGF satisfies

$$G_t = [\mu - (\mu + \beta)x + \beta x^2]G_x + \alpha(x-y)G_y, \quad G(x, y, 0) = y^N. \quad (104)$$

Birth makes infinitely many interior states reachable, but independence has not disappeared. Descendants of distinct exterior particles remain independent, so $G = h^N$, where h is the PGF generated by one exterior particle. What has changed is that h is no longer a finite polynomial.

There is a branch-safe representation of h that is valid for every non-negative choice of the rates. Let $g(x, u)$ be the PGF of a linear birth–death population of age u , started from one interior particle. From [section 2.3](#),

$$g(x, u) = \begin{cases} \frac{\mu(1-x) - (\mu - \beta x)e^{-(\beta-\mu)u}}{\beta(1-x) - (\mu - \beta x)e^{-(\beta-\mu)u}}, & \beta \neq \mu, \\ 1 - \frac{1-x}{1 + \beta u(1-x)}, & \beta = \mu. \end{cases} \quad (105)$$

If the exterior particle has not been absorbed by time t , it contributes y ; if absorption occurs at time $s \leq t$, it initiates an interior process of age $t - s$. Conditioning on the absorption time therefore gives

$$h(x, y, t) = e^{-\alpha t}y + \int_0^t \alpha e^{-\alpha s}g(x, t-s)ds, \quad G(x, y, t) = h(x, y, t)^N. \quad (106)$$

This finite integral is the preferred global formula. It remains regular when $\beta = \mu$, when $\beta = 0$, and at the resonant rate combinations that make individual terms in a special-function expansion singular.

6.5 Characteristic and hypergeometric form

The characteristic system for (104) has the same G , t and y equations as (100), while

$$\frac{dx}{d\tau} = \mu - (\mu + \beta)x + \beta x^2. \quad (107)$$

Assume for the moment that $\beta > 0$ and $b := \beta - \mu \neq 0$. The right-hand side factorises as

$$\beta(x - x_+)(x - x_-), \quad x_+ = 1, \quad x_- = \frac{\mu}{\beta}, \quad (108)$$

and separation gives

$$\frac{x - x_+}{x - x_-} = \frac{\xi - x_+}{\xi - x_-}e^{b\tau}. \quad (109)$$

Write

$$Z = \frac{\xi - x_+}{\xi - x_-}, \quad \zeta(\tau) = Ze^{b\tau}.$$

Solving (109) for x gives

$$x(\tau) = x_+ + (x_+ - x_-)\frac{\zeta(\tau)}{1 - \zeta(\tau)}. \quad (110)$$

The remaining characteristic is

$$\frac{d}{d\tau}(e^{\alpha\tau}y) = \alpha e^{\alpha\tau}x(\tau). \quad (111)$$

Where $|\zeta| < 1$, expand the quotient geometrically and integrate term by term. With

$$\delta = \frac{\alpha}{b}, \quad \Psi(z) = 1 + \frac{\alpha}{\beta} \sum_{n=1}^{\infty} \frac{z^n}{n + \delta}, \quad (112)$$

one obtains $\alpha \int e^{\alpha\tau}x(\tau)d\tau = e^{\alpha\tau}\Psi(\zeta(\tau))$ up to a constant. At the endpoint (x, y, t) , set

$$W = \frac{x - 1}{x - \mu/\beta}. \quad (113)$$

Because $t = -\tau$, the initial characteristic ratio is $Z = We^{bt}$. Solving (111) for the initial coordinate η and using $G = \eta^N$ gives the closed expression

$$G(x, y, t) = \left\{ e^{-\alpha t} [y - \Psi(W)] + \Psi(We^{bt}) \right\}^N. \quad (114)$$

Equation (114) is first obtained in the joint convergence region

$$|W| < 1, \quad |W|e^{bt} < 1, \quad (115)$$

with $n + \delta \neq 0$ for every $n \geq 1$. Within that region it agrees with the integral solution (106), and the identity supplies its analytic continuation wherever a consistent branch is chosen. For numerical work outside (115), especially in the supercritical regime at $x \leq \mu/\beta$, the finite integral is safer: W may cross the standard hypergeometric branch cut even though the physical PGF remains real and analytic.

The series is a Gauss hypergeometric function. Since

$${}_2F_1(1, \delta; \delta + 1; z) = \sum_{n=0}^{\infty} \frac{\delta}{n + \delta} z^n, \quad (116)$$

we may write

$$\Psi(z) = 1 + \frac{\beta - \mu}{\beta} [{}_2F_1(1, \delta; \delta + 1; z) - 1]. \quad (117)$$

This form is compact, but it does not remove the need to specify analytic continuation.

Three checks are independent of that representation. At $t = 0$, the two Ψ terms in (114) cancel and $G = y^N$. At $x = y = 1$, $W = 0$ and $\Psi(0) = 1$, giving $G = 1$. Finally,

$$G(1, y, t) = [1 - e^{-\alpha t} + e^{-\alpha t} y]^N, \quad (118)$$

so $Y_t \sim \text{Binomial}(N, e^{-\alpha t})$, as it must: interior birth and death cannot alter whether an exterior founder has been absorbed.

Two degenerate cases now have a precise status. If $\beta = \mu$, the roots in (108) coalesce and b , δ and W cannot be used; the critical PGF in (105) inserted into (106) is the exact replacement. If

$$\alpha = n(\mu - \beta) \quad \text{for some } n \geq 1, \quad (119)$$

then a denominator $n + \delta$ in (112) vanishes. The singularity is removable in the difference of Ψ terms in (114); its limit contains a factor of t multiplying the resonant exponential. Again, (106) evaluates the solution directly without a parameter limit.

The progression from (91) to (104) is the part that later models reuse. State transitions determine the master equation; the PGF compresses it; characteristics expose the closed form where one exists; and the probabilistic integral keeps the assumptions and exceptional parameter values explicit.

7 What carries forward

The chapter leaves later models with a small set of results that can be reused without rebuilding their probabilistic foundations. For a discrete binary lineage, survival is governed by the exact recursion $S_{n+1} = 2pS_n - pS_n^2$. It separates the three growth regimes, gives the critical tail $S_n \sim 2/n$, and, below criticality, defines the survival amplitude $A(p)$. Conditioning on survival then produces a Yaglom limit with mean $1/A(p)$, while a finite founding cohort changes the chance of reaching late time but not this limiting conditional scale.

The constant $A(p)$ is available at three levels of resolution. Its product and series representations are exact; the bounds and the interval (73) provide certified finite computations; and the asymptotic $A(p) \sim 2(1 - 2p)$ explains the divergence near criticality. The continuous-time constant $A_c(p) = (1 - 2p)/(1 - p)$ is not interchangeable with $A(p)$: their difference records the available surviving states under asynchronous and generational updates. Later applications must therefore choose the clock before importing a conditional population scale.

Catastrophe adds a separate selection mechanism. The weighted recursion (83) shows that conditioning on avoidance of rupture reweights entire population paths, not only their terminal counts. A quasi-stationary result for ordinary branching consequently cannot be transferred to a killed or rupturing process without defining its transient kernel and conditioning event. This point becomes more important, not less, in two-type models, where the catastrophe rate may depend on the composition as well as the total census.

For transfer between compartments, the bivariate PGF keeps the interior and exterior counts distinct. Pure absorption and absorption–death give binomial and trinomial laws, while absorption–birth–death has the global integral solution (106) and, away from critical and resonant parameters, the characteristic form (114). In a multi-compartment extension, the same construction can be applied after a rupture by replacing the single exterior cohort with the released burst-size distribution and assigning one absorption channel to each receiving compartment. Multivariate PGFs then record the resulting joint counts; no new probability principle is required, although the state space and characteristic geometry become larger.

These results also mark the boundary between model and application. The mathematics states what follows from independent reproduction, constant state-dependent hazards and the chosen event ordering. Whether those assumptions are adequate for a particular population, compartment or rupture mechanism must be decided in the relevant application chapter. The quantities to test are now specific: early extinction probabilities, the survival-conditioned scale, catastrophe-time and burst-size laws, absorption times, and the sensitivity of each to distance from criticality.

A A hypergeometric antiderivative

The quotient that appears when the x -characteristic is inserted directly into the y -equation has the form

$$I(\tau) = \int \frac{e^{h\tau}}{a + be^{k\tau}} d\tau. \quad (120)$$

An Euler-integral proof would require a sign condition that excludes the subcritical application. A local power-series proof gives the correct identity first, after which analytic continuation supplies the wider domain.

Proposition A.1. *Suppose $a, h, k \neq 0$ and $h + nk \neq 0$ for every integer $n \geq 0$. On a simply connected domain where a consistent branch is chosen and $a + be^{k\tau} \neq 0$,*

$$I(\tau) = \frac{e^{h\tau}}{ah} {}_2F_1\left(1, \frac{h}{k}; \frac{h}{k} + 1; -\frac{b}{a}e^{k\tau}\right) + C, \quad (121)$$

where the right-hand side is understood by analytic continuation from its power series whenever necessary.

Proof. First suppose $|(b/a)e^{k\tau}| < 1$. The geometric series gives

$$\frac{e^{h\tau}}{a + be^{k\tau}} = \frac{1}{a} \sum_{n=0}^{\infty} \left(-\frac{b}{a}\right)^n e^{(h+nk)\tau}, \quad (122)$$

$$I(\tau) = \frac{1}{a} \sum_{n=0}^{\infty} \left(-\frac{b}{a}\right)^n \frac{e^{(h+nk)\tau}}{h+nk} + C. \quad (123)$$

Put $\nu = h/k$. Since

$${}_2F_1(1, \nu; \nu + 1; z) = \sum_{n=0}^{\infty} \frac{\nu}{\nu + n} z^n,$$

the series in (123) is precisely (121). Both sides are analytic away from their poles and chosen branch cuts, so equality extends throughout any connected domain on which the same continuation is used. $\square$

If $h+n_0k = 0$ for an integer $n_0 \geq 0$, the corresponding term in (123) integrates to $a^{-1}(-b/a)^{n_0}\tau$ rather than an exponential quotient. This is the secular term that appears at the resonance (119); the original finite integral remains regular.

B Recovering state probabilities from a PGF

A PGF is analytic in the open unit disc, or in the unit polydisc for several counts. Its Taylor coefficients are therefore the state probabilities. For the bivariate PGF (88),

$$p_{ij}(t) = \frac{1}{i!j!} \left. \frac{\partial^{i+j} G}{\partial x^i \partial y^j} \right|_{x=y=0}. \quad (124)$$

Analyticity, rather than repeated differentiability alone, is the property that justifies equality with the full Taylor series.

For the finite-state absorption models,

$$G(x, y, t) = [c(t) + f(t)x + g(t)y]^N. \quad (125)$$

Repeated differentiation gives

$$\frac{\partial^{i+j} G}{\partial x^i \partial y^j} = \frac{N!}{(N-i-j)!} f(t)^i g(t)^j [c(t) + f(t)x + g(t)y]^{N-i-j} \quad (126)$$

for $i+j \leq N$, and zero otherwise. Substitution into (124) yields the trinomial law

$$p_{ij}(t) = \frac{N!}{i!j!(N-i-j)!} f(t)^i g(t)^j c(t)^{N-i-j}, \quad i+j \leq N. \quad (127)$$

For pure absorption, $c = 0$, $f = 1 - e^{-\alpha t}$ and $g = e^{-\alpha t}$, so only states with $i+j = N$ have positive probability. For absorption–death, $c = q_d$, $f = q_i$ and $g = q_o$, giving the three independent fates in (98).

The absorption–birth–death PGF is not a finite polynomial in x . Its coefficients may still be obtained from (124); numerically, a stable alternative is the bivariate Cauchy formula

$$p_{ij}(t) = \frac{1}{(2\pi i)^2} \oint_{|x|=r_x} \oint_{|y|=r_y} \frac{G(x, y, t)}{x^{i+1} y^{j+1}} dy dx, \quad 0 < r_x, r_y < 1. \quad (128)$$

Using the integral representation (106) inside these contours avoids the branch bookkeeping of the hypergeometric form.

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